

Assignment ::- 8

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Subject	DBMS

Q1) Create a MySQL database named CollegeDB and create two tables: Course and Student.

Course Table

Column Name	Data Type	Constraint
CourseID	INT	PRIMARY KEY
CourseName	VARCHAR(50)	NOT NULL
Fees	DECIMAL(10,2)	NOT NULL

Student Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
CourseID	INT	FOREIGN KEY
City	VARCHAR(30)	
Marks	INT	
Email	VARCHAR(60)	

Insert at least 4 course records and 10 student records.

-- Q1: Create Database

```
CREATE DATABASE CollegeDB;
```

```
USE CollegeDB;
```

-- Create Course Table

```
CREATE TABLE Course (  
    CourseID INT PRIMARY KEY,  
    CourseName VARCHAR(50) NOT NULL,  
    Fees DECIMAL(10,2) NOT NULL  
);
```

-- Create Student Table

```
CREATE TABLE Student (  
    StudentID INT PRIMARY KEY,
```

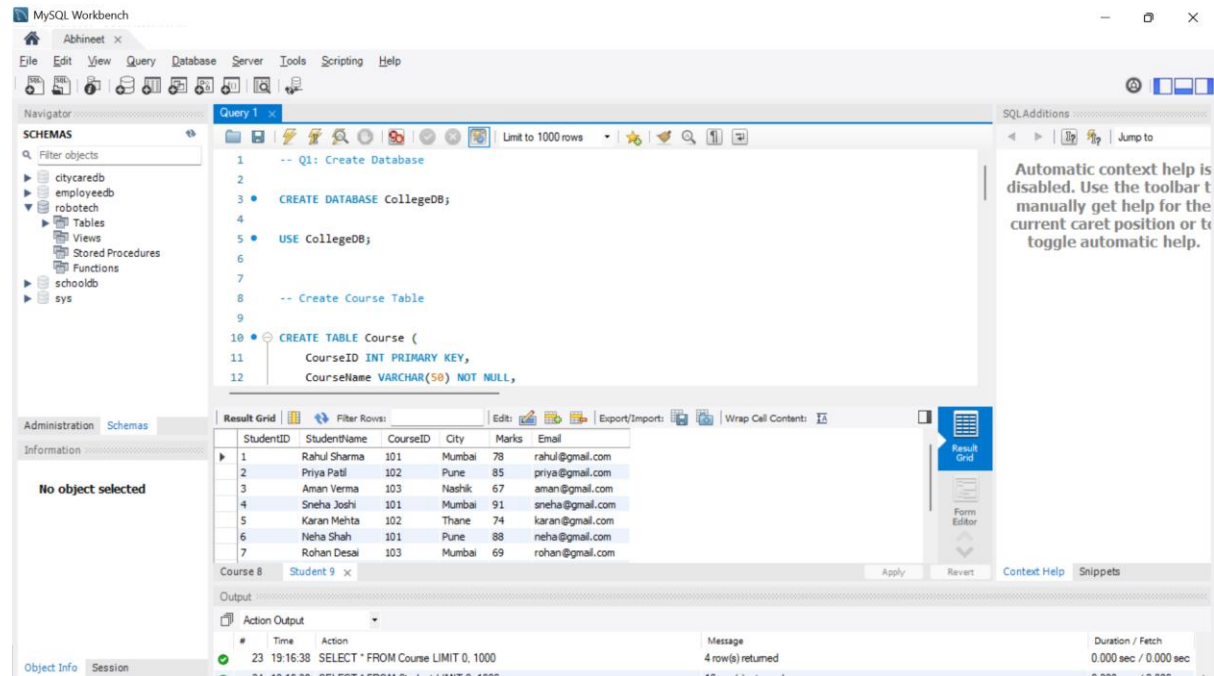
```
StudentName VARCHAR(50) NOT NULL,
CourseID INT,
City VARCHAR(30),
Marks INT,
Email VARCHAR(60),
FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
);
-- Insert records into Course
INSERT INTO Course (CourseID, CourseName, Fees)
VALUES
(101, 'BSc IT', 45000.00),
(102, 'BCA', 40000.00),
(103, 'BSc CS', 42000.00),
(104, 'Data Science', 55000.00);
-- Insert records into Student
INSERT INTO Student
(StudentID, StudentName, CourseID, City, Marks, Email)
VALUES
(1, 'Rahul Sharma', 101, 'Mumbai', 78, 'rahul@gmail.com'),
(2, 'Priya Patil', 102, 'Pune', 85, 'priya@gmail.com'),
(3, 'Aman Verma', 103, 'Nashik', 67, 'aman@gmail.com'),
(4, 'Sneha Joshi', 101, 'Mumbai', 91, 'sneha@gmail.com'),
(5, 'Karan Mehta', 102, 'Thane', 74, 'karan@gmail.com'),
(6, 'Neha Shah', 101, 'Pune', 88, 'neha@gmail.com'),
(7, 'Rohan Desai', 103, 'Mumbai', 69, 'rohan@gmail.com'),
(8, 'Meera Kulkarni', 102, 'Nashik', 82, 'meera@gmail.com'),
(9, 'Arjun Nair', 101, 'Thane', 59, 'arjun@gmail.com'),
(10, 'Divya More', 102, 'Mumbai', 93, 'divya@gmail.com');
```

-- Display Course Table

SELECT * FROM Course;

-- Display Student Table

SELECT * FROM Student;



Q2) Create a view named StudentBasicView to display only selected student details.

The view should contain:

Column
StudentID
StudentName
Column
City
Marks

CREATE VIEW StudentBasicView AS

SELECT

StudentID,

StudentName,

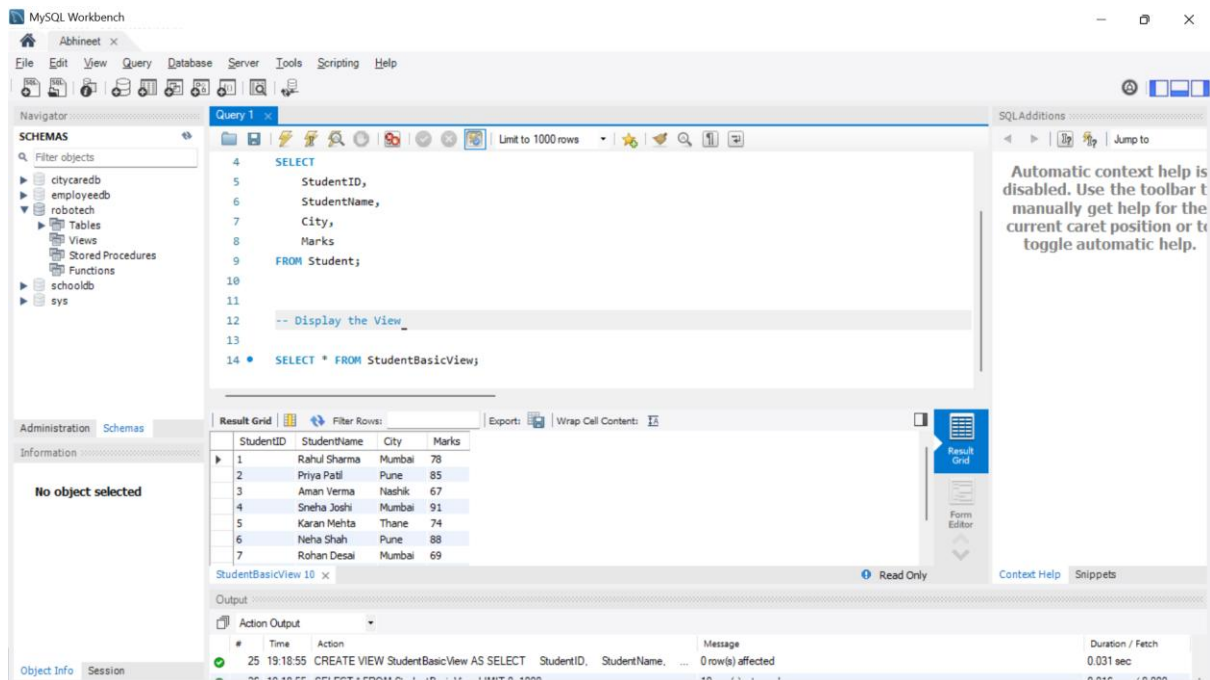
City,

Marks

FROM Student

-- Display the View

SELECT * FROM StudentBasicView;



Q3) Create a view named StudentCourseView using Student and Course tables.

The view should display:

Column
StudentID
StudentName
City
Marks
CourseName
Fees

CREATE VIEW StudentCourseView AS

SELECT

Student.StudentID,

Student.StudentName,

Student.City,

Student.Marks,

Course.CourseName,

Course.Fees

FROM Student

INNER JOIN Course

ON Student.CourseID = Course.CourseID;

-- Display the View

SELECT * FROM StudentCourseView;

The screenshot shows the MySQL Workbench interface. The 'Query Editor' window contains the following SQL code:

```
8 Student.Marks,  
9 Course.CourseName,  
10 Course.Fees  
11 FROM Student  
12 INNER JOIN Course  
13 ON Student.CourseID = Course.CourseID;  
14  
15  
16 -- Display the View  
17  
18 SELECT * FROM StudentCourseView;
```

The 'Result Grid' window displays the output of the query, showing a table with 6 columns: StudentID, StudentName, City, Marks, CourseName, and Fees. The data is as follows:

StudentID	StudentName	City	Marks	CourseName	Fees
1	Rahul Sharma	Mumbai	78	BSc IT	45000.00
4	Sneha Joshi	Mumbai	91	BSc IT	45000.00
6	Neha Shah	Pune	88	BSc IT	45000.00
9	Arjun Nair	Thane	59	BSc IT	45000.00
2	Priya Patil	Pune	85	BCA	40000.00
5	Karan Mehta	Thane	74	BCA	40000.00
8	Meera Kulkarni	Nashik	82	BCA	40000.00

The 'Output' window shows the execution details:

#	Time	Action	Message	Duration / Fetch
27	19:19:52	CREATE VIEW StudentCourseView AS SELECT Student.StudentID, Student...	0 row(s) affected	0.031 sec

Q4) Write a MySQL query using StudentCourseView to display students from Mumbai.

The output should contain:

Column
StudentName
City
CourseName
Marks

SELECT

StudentName,

City,

CourseName,

Marks

FROM StudentCourseView

WHERE City = 'Mumbai';

The screenshot shows the MySQL Workbench interface. The 'Query Editor' window contains the following SQL query:

```
-- Q4: Display students from Mumbai  
  
SELECT  
    StudentName,  
    City,  
    CourseName,  
    Marks  
FROM StudentCourseView  
WHERE City = 'Mumbai';
```

The 'Result Grid' window displays the results of the query:

StudentName	City	CourseName	Marks
Rahul Sharma	Mumbai	BSc IT	78
Sneha Joshi	Mumbai	BSc IT	91
Rohan Desai	Mumbai	BSc CS	69
Divya More	Mumbai	BCA	93

The 'Output' window shows the execution details:

#	Time	Action	Message	Duration / Fetch
28	19:19:52	SELECT * FROM StudentCourseView LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

Q5) Write a MySQL query using StudentCourseView to display students sorted by marks in descending order.

The output should contain:

Column
StudentName
CourseName
Marks

SELECT

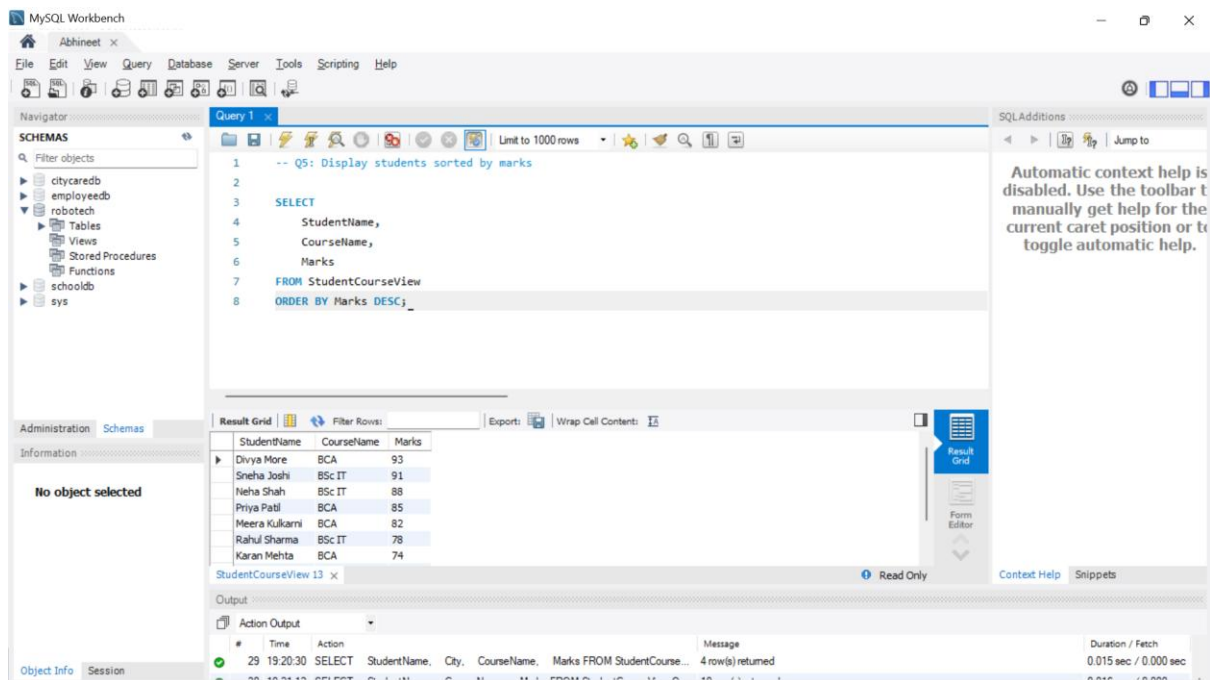
StudentName,

CourseName,

Marks

FROM StudentCourseView

ORDER BY Marks DESC;



Q6) Create a report view named *ManagementReportView*.

The view should display course-wise summary:

Column
CourseName
TotalStudents
AverageMarks
LowestMarks
HighestMarks

Also write one line explaining why such a view is useful in reporting

CREATE VIEW ManagementReportView AS

SELECT

Course.CourseName,

COUNT(Student.StudentID) AS TotalStudents,

AVG(Student.Marks) AS AverageMarks,

MIN(Student.Marks) AS LowestMarks,

MAX(Student.Marks) AS HighestMarks

FROM Course

LEFT JOIN Student

ON Course.CourseID = Student.CourseID

GROUP BY Course.CourseName;

-- Display the Report

SELECT * FROM ManagementReportView;

The screenshot shows the MySQL Workbench interface. The main window displays a SQL query in the 'Query Editor' tab. The query is as follows:

```
8      MIN(Student.Marks) AS LowestMarks,  
9      MAX(Student.Marks) AS HighestMarks  
10     FROM Course  
11     LEFT JOIN Student  
12     ON Course.CourseID = Student.CourseID  
13     GROUP BY Course.CourseName;  
14  
15  
16     -- Display the Report  
17  
18     SELECT * FROM ManagementReportView;
```

The 'Result Grid' tab shows the output of the query. The table has the following columns: CourseName, TotalStudents, AverageMarks, LowestMarks, and HighestMarks. The data is as follows:

CourseName	TotalStudents	AverageMarks	LowestMarks	HighestMarks
BSc IT	4	79.0000	59	91
BCA	4	83.5000	74	93
BSc CS	2	68.0000	67	69
Data Science	0	NULL	NULL	NULL

The 'Output' tab shows the execution log. The first entry is:

```
31 19:21:51 CREATE VIEW ManagementReportView AS SELECT Course.CourseName, C... 0 row(s) affected 0.031 sec
```